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Configuring SSH Remote Access on Ubuntu Server

Objective

Set up an Ubuntu Server with SSH (Secure Shell) remote access and connect to it from a host desktop machine using Bridged Network mode.

Operating System Installation

Install Ubuntu Server

1. Start the VM and boot from the Ubuntu Server ISO
2. Follow the installation wizard:
 - Profile setup: Create a user account (e.g., server / server123)
 - Install OpenSSH server: **Check this option** (Important!)
 - Selected software: **None** (Standard Ubuntu Server)
3. Wait for installation to complete and reboot

Network Configuration

Configure Network Mode

For VirtualBox:

1. Power off the VM (if running)
2. Go to **Settings** → **Network**
3. **Adapter 1**: Enable Network Adapter
4. **Attached to**: Select **Bridged Adapter**
5. **Name**: Select your host's physical network adapter (Wi-Fi/Ethernet)
6. Click **OK**

For VMware:

1. Power off the VM
2. Go to **VM Settings** → **Network Adapter**

3. Select **Bridged** mode

4. Click **OK**

Install/Verify SSH Server

Server: Install SSH on the server (if not already installed):

- `sudo systemctl status ssh` – for checking the SSH status.
- `sudo apt update` – updating the your server.
- `sudo apt install openssh-server -y` –installing the SSH.
- `sudo systemctl enable ssh` – enabling the SSH to your unit. (SERVER/DESKTOP)
- `sudo systemctl start ssh` – starting the SSH to your unit. (SERVER/DESKTOP)

Bridged makes the server behave like a separate PC in your LAN.

1. Find the Server's IP

Inside the VM, run:

```
Sample@UBUNTU-DESKTOP:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:9b:8c:ca brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.4/24 brd 192.168.1.255 scope global dynamic noprefixroute enp0s3
        valid_lft 83602sec preferred_lft 83602sec
    inet6 2001:4451:b00:2000:d76c:1af2:a10c:f360/64 scope global temporary dynamic
        valid_lft 179068sec preferred_lft 83386sec
    inet6 2001:4451:b00:2000:a00:27ff:fe9b:8cca/64 scope global dynamic mngtproto kernel_rdr proto kernel_rdr
        valid_lft 179068sec preferred_lft 92668sec
    inet6 fe80::a00:27ff:fe9b:8cca/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
Sample@UBUNTU-DESKTOP:~$
```

From the desktop, connect to the server: `ssh username@192.168.1.120`

2. Connect from Desktop

From your host desktop:

```
Sample@UBUNTU-DESKTOP:~$ ssh Server@192.168.1.14
Server@192.168.1.14's password:
Welcome to Ubuntu 24.04.1 LTS (GNU/Linux 6.8.0-79-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Mon Sep  8 11:24:38 PM UTC 2025

System load:          0.0
Usage of /:           5.1% of 48.91GB
Memory usage:         5%
Swap usage:           0%
Processes:            139
Users logged in:      1
IPv4 address for enp0s3: 192.168.1.14
IPv6 address for enp0s3: 2001:4451:b00:2000:a00:27ff:fe57:bc4a

Expanded Security Maintenance for Applications is not enabled.

136 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Mon Sep  8 22:52:07 2025 from 192.168.1.4
Server@UBUNTU-SERVER:~$
```

Submission Requirements

1. Screenshots (labeled clearly):

- VM network settings showing Bridged Adapter
- Server IP address (output of `ip a` or `hostname -I`)
- SSH service status (`sudo systemctl status ssh`)
- Successful SSH connection from host
- Verification commands executed inside VM

2. Submission Format:

- Single PDF file
- Include name, student ID, and date
- All screenshots with brief explanation
- Upload to your portfolio website.

UBUNTU SERVER SSH REMOTE ACCESS LAB

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Course/Year/Section: BSIT 3B

Date Submitted: 09/02/26

Instructor: Johneros Puyo

1 OBJECTIVES (5 POINTS)

Check all that apply:

- ✓ ☐ Create and configure a Virtual Machine with Ubuntu Server
- ✓ ☐ Configure Bridged Network Adapter for network access
- ✓ ☐ Determine and verify the server's IP address
- ✓ ☐ Install OpenSSH Server and enable the service
- ✓ ☐ Establish successful SSH connection from host machine
- ✓ ☐ Execute verification commands to confirm remote access

2 HARDWARE & SOFTWARE REQUIREMENTS (5 POINTS)

Item	Specification
Host Operating System	Ubuntu Server 26
Virtualization Software (VirtualBox/VMware)	Oracle VM VirtualBox 7.0
Ubuntu Server Version	26.04
VM Name	Jimeno
RAM Allocated	4096 MB
Storage Allocated	50GB
Network Mode	Bridge Adapter
SSH Client Used	Secure shell client

3 VM CONFIGURATION

Setting	Value
VM Name	UBUNTU-SERVER
OS Type	Ubuntu (64-bit)

Memory	4096 MB
Hard Disk	50 GB
Network Adapter	Bridged

4. NETWORK CONFIGURATION

Commands Executed:

Server IP Address: 192.168.137.32

Network Interface: Bridged Network

6. REMOTE CONNECTION

Connection Details:

Parameter	Value
Server IP	192.168.137.32
Username	kent
SSH Port	ssh kent@192.168.137.32



Create Virtual Machine

?

×

▼ Name and Operating System

Name: 

Folder: ▼

ISO Image: ▼

Edition: ▼

Type: ▼ 

Version: ▼

☐ Skip Unattended Installation

> Unattended Install

> Hardware

> Hard Disk

Help

Guided Mode

Back

Finish

Cancel

Create Virtual Machine

> Name and Operating System

> Unattended Install

< Hardware

Base Memory:

4 MB

16384 MB

4096 MB

Processors:

1 CPU

4

Holds the number of virtual CPUs in the virtual machine. You need hardware virtualization support in your host operating system to use more than one virtual CPU.

☐ Enable EFI (special OSes only)

> Hard Disk

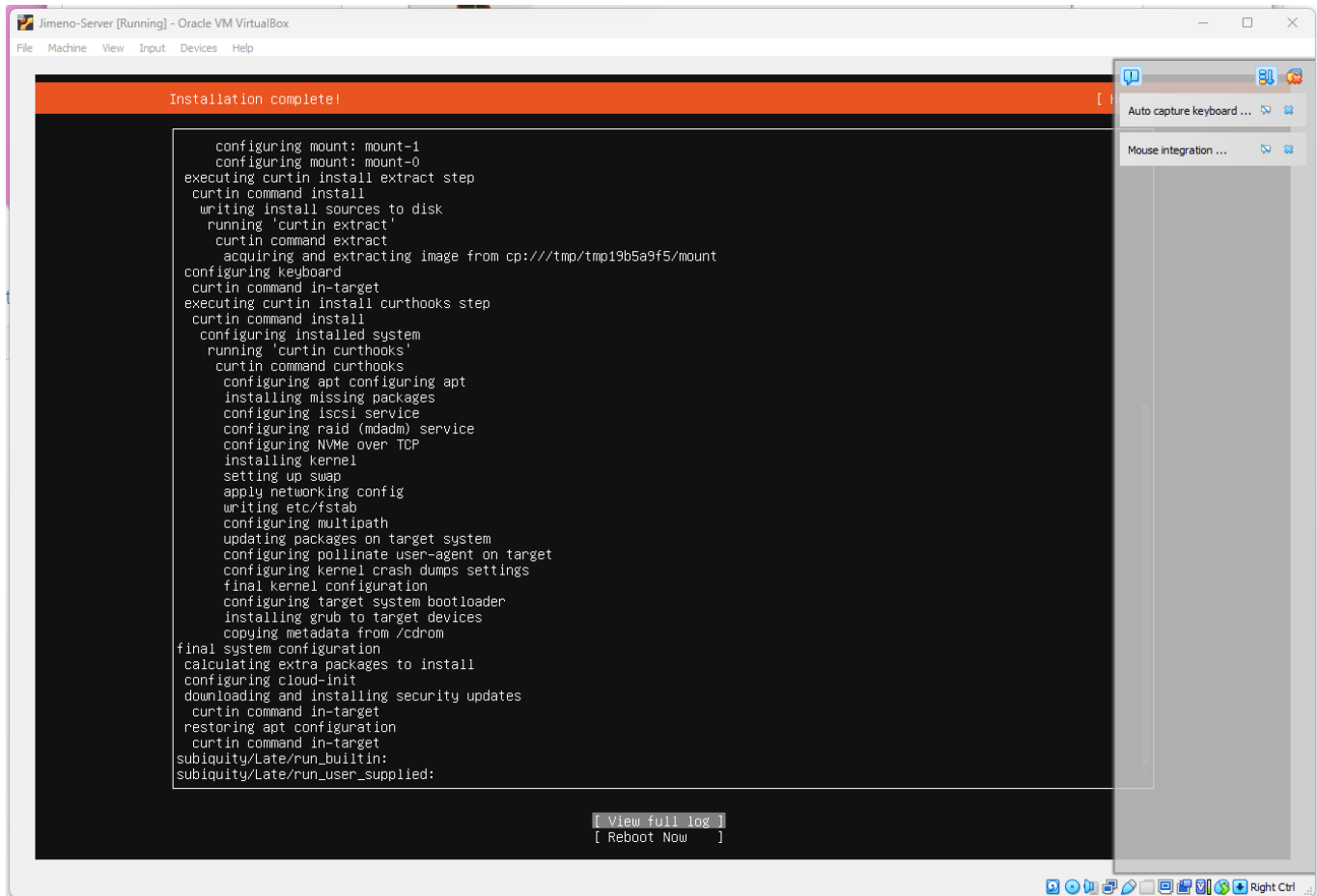
Help

Guided Mode

Back

Finish

Cancel



```
jimeno login: kentruess
Password:
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-30-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Wed Sep  2 06:22:49 AM UTC 2026

System load:  0.5               Processes:    154
Usage of /:   29.8% of 23.45GB   Users logged in: 0
Memory usage: 7%               IPv4 address for enp0s3: 10.0.2.15
Swap usage:  0%

Expanded Security Maintenance for Applications is not enabled.

53 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
```

```

Preparing to unpack .../libwrap0_7.6.q-36build2_amd64.deb...
Unpacking libwrap0:amd64 (7.6.q-36build2)...
Selecting previously unselected package ncurses-term.
Preparing to unpack .../ncurses-term_6.6+20251231-1_all.deb...
Unpacking ncurses-term (6.6+20251231-1)...
Selecting previously unselected package openssh-sftp-server.
Preparing to unpack .../openssh-sftp-server_1%3a10.2p1-2ubuntu3.5_amd64.deb...
Unpacking openssh-sftp-server (1:10.2p1-2ubuntu3.5)...
Selecting previously unselected package openssh-server.
Preparing to unpack .../openssh-server_1%3a10.2p1-2ubuntu3.5_amd64.deb...
Unpacking openssh-server (1:10.2p1-2ubuntu3.5)...
Selecting previously unselected package ssh-import-id.
Preparing to unpack .../ssh-import-id_5.11-0ubuntu4_all.deb...
Unpacking ssh-import-id (5.11-0ubuntu4)...
Setting up openssh-sftp-server (1:10.2p1-2ubuntu3.5)...
Setting up ssh-import-id (5.11-0ubuntu4)...
Setting up libwrap0:amd64 (7.6.q-36build2)...
Setting up ncurses-term (6.6+20251231-1)...
Setting up openssh-server (1:10.2p1-2ubuntu3.5)...
Creating config file /etc/ssh/sshd_config with new version
Creating user 'sshd' (sshd user) with UID 983 and GID 65534.
Created symlink '/etc/systemd/system/sockets.target.wants/ssh.socket' → '/usr/lib/systemd/system/ssh.socket'.
Created symlink '/etc/systemd/system/ssh.service.requires/ssh.socket' → '/usr/lib/systemd/system/ssh.socket'.
Created symlink '/etc/systemd/system/ssh.service.wants/sshd-keygen.service' → '/usr/lib/systemd/system/sshd-keygen.service'.
Created symlink '/etc/systemd/system/sshd.service.wants/sshd-keygen.service' → '/usr/lib/systemd/system/sshd-keygen.service'.
Created symlink '/etc/systemd/system/sshd@.service.wants/sshd-keygen.service' → '/usr/lib/systemd/system/sshd-keygen.service'.
Created symlink '/etc/systemd/system/ssh.socket.wants/sshd-keygen.service' → '/usr/lib/systemd/system/sshd-keygen.service'.
Processing triggers for ufw (0.36.2-9build1)...
Processing triggers for man-db (2.13.1-1build1)...
Processing triggers for libc-bin (2.43-2ubuntu2.3)...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
kentruss@jimeno:~$ sudo apt install openssh-server-y
Error: Unable to locate package openssh-server-y
kentruss@jimeno:~$ sudo systemctl enable ssh
Synchronizing state of ssh.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable ssh
Created symlink '/etc/systemd/system/sshd.service' → '/usr/lib/systemd/system/ssh.service'.
Created symlink '/etc/systemd/system/multi-user.target.wants/ssh.service' → '/usr/lib/systemd/system/ssh.service'.

kentruss@jimeno:~$ sudo systemctl status ssh
* ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: inactive (dead)
TriggeredBy: ● ssh.socket
   Docs: man:sshd(8)
        man:sshd_config(5)

```

```

kentruuss@jimeno:~$ sudo systemctl status ssh
[sudo: authenticate] Password:
• ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-09-02 06:37:25 UTC; 50s ago
   Invocation: bb951facdef343c7af8b3eb993c0ce68
   TriggeredBy: • ssh.socket
     Docs: man:sshd(8)
           man:sshd_config(5)
   Process: 1284 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 1292 (sshd)
     Tasks: 1 (limit: 1683)
    Memory: 1.6M (peak: 2.4M)
       CPU: 32ms
    CGroup: /system.slice/ssh.service
            └─1292 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Sep 02 06:37:25 jimeno systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Sep 02 06:37:25 jimeno sshd[1292]: Server listening on 0.0.0.0 port 22.
Sep 02 06:37:25 jimeno sshd[1292]: Server listening on :: port 22.
Sep 02 06:37:25 jimeno systemd[1]: Started ssh.service - OpenBSD Secure Shell server.

kentruuss@jimeno:~$ sudo apt update
Ign:1 http://security.ubuntu.com/ubuntu resolute-security InRelease
Ign:2 http://ph.archive.ubuntu.com/ubuntu resolute InRelease
Ign:3 http://ph.archive.ubuntu.com/ubuntu resolute-updates InRelease
Ign:4 http://ph.archive.ubuntu.com/ubuntu resolute-backports InRelease
Ign:4 http://ph.archive.ubuntu.com/ubuntu resolute-backports InRelease
Ign:1 http://security.ubuntu.com/ubuntu resolute-security InRelease
Ign:3 http://ph.archive.ubuntu.com/ubuntu resolute-updates InRelease
Ign:2 http://ph.archive.ubuntu.com/ubuntu resolute InRelease
Ign:2 http://ph.archive.ubuntu.com/ubuntu resolute InRelease
Ign:1 http://security.ubuntu.com/ubuntu resolute-security InRelease
Ign:3 http://ph.archive.ubuntu.com/ubuntu resolute-updates InRelease
Ign:4 http://ph.archive.ubuntu.com/ubuntu resolute-backports InRelease
Err:4 http://ph.archive.ubuntu.com/ubuntu resolute-backports InRelease
   Temporary failure resolving 'ph.archive.ubuntu.com'
Err:1 http://security.ubuntu.com/ubuntu resolute-security InRelease
   Temporary failure resolving 'security.ubuntu.com'
Err:3 http://ph.archive.ubuntu.com/ubuntu resolute-updates InRelease
   Temporary failure resolving 'ph.archive.ubuntu.com'
Err:2 http://ph.archive.ubuntu.com/ubuntu resolute InRelease
   Temporary failure resolving 'ph.archive.ubuntu.com'
55 packages can be upgraded. Run 'apt list --upgradable' to see them.
Warning: Failed to fetch http://ph.archive.ubuntu.com/ubuntu/dists/resolute/InRelease Temporary failure resolving 'ph.archive.ubuntu.com'
Warning: Failed to fetch http://ph.archive.ubuntu.com/ubuntu/dists/resolute-updates/InRelease Temporary failure resolving 'ph.archive.ubuntu.com'
Warning: Failed to fetch http://ph.archive.ubuntu.com/ubuntu/dists/resolute-backports/InRelease Temporary failure resolving 'ph.archive.ubuntu.com'
Warning: Failed to fetch http://security.ubuntu.com/ubuntu/dists/resolute-security/InRelease Temporary failure resolving 'security.ubuntu.com'
Warning: Some index files failed to download. They have been ignored, or old ones used instead.

kentruuss@jimeno:~$ sudo apt install openssh-server -y
openssh-server is already the newest version (1:10.2p1-2ubuntu3.5).
The following packages were automatically installed and are no longer required:
  linux-headers-7.0.0-14      linux-image-unsigned-7.0.0-14-generic  linux-modules-7.0.0-14-generic  linux-tools-7.0.0-14-generic
  linux-headers-7.0.0-14-generic  linux-main-modules-zfs-7.0.0-14-generic  linux-tools-7.0.0-14
Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 55

kentruuss@jimeno:~$ sudo systemctl enable ssh
Synchronizing state of ssh.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable ssh

```

```
Jimeneserver [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

Ubuntu 26.04 LTS kent tty1

kent login: kent
Password:
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-30-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Wed Sep  2 09:40:33 UTC 2026

System load: 0.0           Memory usage: 7%      Processes:   158
Usage of /: 30.0% of 23.45GB Swap usage:   0%      Users logged in: 0

Expanded Security Maintenance for Applications is not enabled.

53 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

kent@kent:~$ sudo apt update
(sudo: authenticate) Password:
Hit:1 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:2 http://ph.archive.ubuntu.com/ubuntu resolute InRelease
Hit:3 http://ph.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:4 http://ph.archive.ubuntu.com/ubuntu resolute-backports InRelease
55 packages can be upgraded. Run 'apt list --upgradable' to see them.
kent@kent:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:5f:17:42 brd ff:ff:ff:ff:ff:ff
    altname enx0800275f1742
    inet 192.168.137.32/24 metric 1024 brd 192.168.137.255 scope global dynamic enp0s8
        valid_lft 604785sec preferred_lft 604785sec
    inet6 fe80::a00:27ff:fe5f:1742/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
kent@kent:~$
```

```
Jimeneserver [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

Ubuntu 26.04 LTS kent tty1

kent login: kent
Password:
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-30-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
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Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

kent@kent:~$ sudo apt update
(sudo: authenticate) Password:
Hit:1 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:2 http://ph.archive.ubuntu.com/ubuntu resolute InRelease
Hit:3 http://ph.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:4 http://ph.archive.ubuntu.com/ubuntu resolute-backports InRelease
55 packages can be upgraded. Run 'apt list --upgradable' to see them.
kent@kent:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:5f:17:42 brd ff:ff:ff:ff:ff:ff
    altname enx0800275f1742
    inet 192.168.137.32/24 metric 1024 brd 192.168.137.255 scope global dynamic enp0s8
        valid_lft 604785sec preferred_lft 604785sec
    inet6 fe80::a00:27ff:fe5f:1742/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
kent@kent:~$
```

